A verifiable nonlinear sufficient-condition theorem for discounted continuous-time turnpikes

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Abstract

We give sufficient conditions, in one class of systems, for the discount-dependent reference and the unweighted measure-turnpike bound of C61-93. For finite-dimensional systems $\dot{x} = g(x) + Bu$ with a nonlinear contractive drift g, a constant matrix B, a state boundary that is strictly inward for every control, compact convex control constraints, a strongly convex control cost, and a possibly nonconvex state cost, explicit bounds on the problem data construct a unique discount-dependent stationary reference at every discount satisfying a quantitative small-gain condition. At each such discount, every finite-horizon optimizer satisfies an unweighted exponential turnpike estimate, uniformly over the stated initial-state and terminal-cost family. We give the constants and rates as functions of the discount and explicit conditions for uniformity and reference continuity as the discount tends to zero. The proof permits active control constraints and derives optimality conditions by admissible first variations. The statement is sufficient, not a necessary-and-sufficient classification of all nonlinear systems.

1 Scope

The canonical C61-93 statement asks for structural assumptions yielding a discount-dependent reference, an *unweighted* measure-turnpike bound uniform in the horizon and in a specified data family, and quantitative dependence on the discount, including sufficient conditions for uniformity as the discount tends to zero. It expressly does not assert a universal turnpike under continuity, compactness, Lipschitz dynamics, and attainment alone. The source [1], Section 5, identifies continuous-time discounting as a research direction; Section 2 explains the horizon and initial-condition uniformity in the definition.

The theorem below supplies one directly checkable sufficient structural regime and proves these conclusions in that regime. The drift and costs can be nonlinear, the dimensions are arbitrary, and the control constraint may be active. No reference, solution of a canonical boundary-value problem, dissipativity inequality along unknown optimizers, or turnpike estimate is assumed. The result is not a necessary-and-sufficient classification of every nonlinear optimal-control problem.

Endpoint and terminal-cost family. Initial states range over a fixed compact set X_0 strictly inside the state ball X below. The terminal state is free, subject only to the state constraint. Terminal costs range over a family $\mathcal{F}$ with uniform bounds on their values *and their first derivatives*. Thus the identified structural hypothesis is stronger than merely bounded terminal values. Prescribed terminal-state constraints are not part of the displayed C61-93 problem or of the family treated here.

2 Data assumptions and explicit constants

Let the state dimension be $d \geq 1$ and the control dimension be $r \geq 1$. Use Euclidean vector norms and their induced matrix norms. Let

$$X = \{x \in \mathbb{R}^d : |x| \leq R_X\}, \qquad R_X > 0,$$

and let $U \subset \mathbb{R}^r$ be nonempty, compact, and convex. Fix a nonempty compact set $X_0 \subset \operatorname{int} X$. Consider

$$\min_{u(t)\in U} \left\{ \int_0^T e^{-\rho t}\big(L(x(t)) + Q(u(t))\big)\, dt + e^{-\rho T}\Phi(x(T)) \right\}, \qquad \dot{x} = g(x) + Bu, \quad x(0) = x_0, \tag{1}$$

with $x(t) \in X$, arbitrary $T > 0$, $x_0 \in X_0$, and $\Phi \in \mathcal{F}$. Here $\rho \geq 0$; the positive values are the discounted problems and $\rho = 0$ is included to make the limiting assertions precise.

Assume the following inequalities hold with finite constants $a, m, \eta > 0$ and $K, M, G, H, F \geq 0$:

1. g is C^2 on a neighborhood of X, B is a constant real matrix, and, for all $x, y \in X$,

$$\operatorname{sym} Dg(x) \preceq -aI, \tag{2}$$
$$\|Dg(x) - Dg(y)\| \leq K|x - y|. \tag{3}$$

2. The state boundary is uniformly strictly inward for *every* control:

$$x \cdot (g(x) + Bu) \leq -\eta \qquad (|x| = R_X,\ u \in U). \tag{4}$$

3. L is C^1 on a neighborhood of X and

$$|\nabla L(x)| \leq G, \qquad |\nabla L(x) - \nabla L(y)| \leq M|x - y| \quad (x, y \in X). \tag{5}$$

   Convexity of L is not required.

4. Q is C^1 on a neighborhood of U and is m-strongly convex on U, in the explicit sense

$$Q(v) \geq Q(u) + \nabla Q(u) \cdot (v - u) + \frac{m}{2}|v - u|^2 \qquad (u, v \in U). \tag{6}$$

5. Each $\Phi \in \mathcal{F}$ is C^1 on a neighborhood of X, with

$$\sup_{\Phi\in\mathcal{F}} \sup_{x\in X} |\Phi(x)| \leq F, \qquad \sup_{\Phi\in\mathcal{F}} \sup_{x\in X} |\nabla\Phi(x)| \leq H. \tag{7}$$

   Convexity of the terminal costs is not required.

These are bounds on the problem data, independent of unknown optimizers and of the horizon. The strict inward condition can, for example, be checked by bounding the maximum of $x\cdot(g(x)+Bu)$ on the compact boundary and control set.

Write $b = \|B\|$, $k = b^2/m$, $D = \operatorname{diam} X = 2R_X$, and, for a given discount, define

$$M_\rho = M + \frac{KG}{a + \rho}, \qquad c_\rho = kM_\rho, \qquad \kappa_\rho = \frac{c_\rho}{a(a + \rho)}. \tag{8}$$

The quantitative small-gain condition at discount ρ is

$$\kappa_\rho < 1. \tag{9}$$

For this discount put

$$\gamma_*(\rho) = a + \frac{\rho}{2} - \sqrt{\frac{\rho^2}{4} + c_\rho} > 0. \tag{10}$$

Choose any $0 < \gamma < \gamma_*(\rho)$ and define

$$\theta_{\rho,\gamma} = \frac{c_\rho}{(a-\gamma)(a+\rho-\gamma)} < 1, \qquad H_\rho = H + \frac{G}{a+\rho}, \tag{11}$$

$$A_{\rho,\gamma} = \frac{D + \dfrac{kH_\rho}{a-\gamma}}{1-\theta_{\rho,\gamma}}, \qquad P_{\rho,\gamma} = H_\rho + \frac{M_\rho A_{\rho,\gamma}}{a+\rho-\gamma}, \tag{12}$$

$$C_{\rho,\gamma} = A_{\rho,\gamma} + \frac{b}{m} P_{\rho,\gamma}. \tag{13}$$

Here $\gamma < a$ follows from (10); all denominators are positive.

Theorem 1 (Global discounted turnpike under data small gain)**.** *Under the preceding data assumptions, every problem* (1) *has an optimal control. For every discount satisfying* (9)*, a unique stationary current-value reference* $(\bar{x}_\rho, \bar{u}_\rho, \bar{p}_\rho)$ *with Hamiltonian-minimizing control is constructed from the data by the contraction specified below. For* every *optimal pair and every* $T > 0$*,* $x_0 \in X_0$*,* $\Phi \in \mathcal{F}$*, one has*

$$|x_T(t) - \bar{x}_\rho| + |u_T(t) - \bar{u}_\rho| \le C_{\rho,\gamma}\left(e^{-\gamma t} + e^{-\gamma(T-t)}\right) \quad \text{for almost every } t \in [0,T]. \tag{14}$$

The optimal control has a continuous representative for which the inequality holds everywhere. In particular, for every $\varepsilon > 0$*,*

$$\mathrm{Leb}\{t \in [0,T] : |x_T(t) - \bar{x}_\rho| + |u_T(t) - \bar{u}_\rho| > \varepsilon\} \le \frac{2}{\gamma} \log_+\left(\frac{2C_{\rho,\gamma}}{\varepsilon}\right), \tag{15}$$

where $\log_+ s = \max(0, \log s)$*. The constants are independent of* T*,* x_0*, and the chosen terminal cost in the specified family.*

If $c_0 < a^2$*, choose any fixed*

$$0 < \gamma < a - \sqrt{c_0}. \tag{16}$$

Then all discounts $\rho \ge 0$ *satisfy the theorem with this same rate and with* $C_{\rho,\gamma} \le C_{0,\gamma}$*. Thus both the exponential and the unweighted measure estimates are uniform as* $\rho \downarrow 0$*. The reference is Lipschitz in* ρ *in this uniform regime, with explicit constants given in Proposition 6 below.*

3 Construction of the reference and existence of optimizers

Lemma 2 (Invariance, constant-control equilibria, and feedback)**.** *Every measurable* U*-valued control starting in* X_0 *produces a trajectory strictly inside* X *for all finite times. For each constant* $u \in U$*, there is a unique* $z(u) \in \operatorname{int} X$ *satisfying* $g(z(u)) + Bu = 0$*, and*

$$|z(u) - z(v)| \le \frac{b}{a}|u - v|. \tag{17}$$

The map

$$\mathcal{K}(p) = \arg\min_{u \in U}\{Q(u) + p \cdot Bu\} \tag{18}$$

is single valued and satisfies

$$|\mathcal{K}(p) - \mathcal{K}(q)| \le \frac{b}{m}|p - q|. \tag{19}$$

Proof. Compactness and continuity in (4) provide a boundary shell on which $x \cdot (g(x) + Bu) \leq -\eta/2$ uniformly over $u \in U$. The absolutely continuous function $|x(t)|^2$ therefore has strictly negative derivative almost everywhere while it lies in that shell. It cannot cross the shell outward. Starting in the fixed compact $X_0 \subset \operatorname{int} X$ gives a positive uniform distance from the boundary for all such controls. Local existence and uniqueness of the controlled ODE, followed by compact continuation, give global existence. This also proves that every convex control variation is admissible; state-constraint multipliers will not be needed.

For the same constant control, (2) and convexity of X give

$$\frac{d}{dt}|x(t) - y(t)|^2 \leq -2a|x(t) - y(t)|^2.$$

The time-one autonomous flow is therefore a strict contraction of the complete space X into itself and has a unique fixed point z. For every $s \geq 0$, the time-s image of z is again a fixed point of the time-one flow, by autonomy and commutation of the flow maps. Uniqueness forces that image to equal z, so z is an equilibrium. Strict inwardness places it in the interior. Equilibria for controls u, v satisfy

$$a|z(u) - z(v)|^2 \leq B(u - v) \cdot (z(u) - z(v)),$$

which implies (17).

The minimum in (18) exists by compactness and is unique by strong convexity. Its first-order variational inequality is

$$[\nabla Q(\mathcal{K}(p)) + B^T p] \cdot [w - \mathcal{K}(p)] \geq 0 \qquad (w \in U).$$

Use $w = \mathcal{K}(q)$ in this inequality and the reverse choice in the corresponding inequality for q. Strong monotonicity of ∇Q then yields $m|\mathcal{K}(p) - \mathcal{K}(q)|^2 \leq b|p - q||\mathcal{K}(p) - \mathcal{K}(q)|$, proving the Lipschitz bound. This argument includes active boundary controls. □

For $x \in X$ define

$$\mathcal{P}_\rho(x) = (\rho I - Dg(x)^T)^{-1} \nabla L(x). \tag{20}$$

The inverse exists: the symmetric part of the matrix in parentheses is at least $(a + \rho)I$. In particular,

$$|\mathcal{P}_\rho(x)| \leq \frac{G}{a + \rho}, \qquad |\mathcal{P}_\rho(x) - \mathcal{P}_\rho(y)| \leq \frac{M_\rho}{a + \rho}|x - y|. \tag{21}$$

For the second inequality, multiply the difference by $\rho I - Dg(x)^T$. The result is

$$\nabla L(x) - \nabla L(y) + (Dg(x) - Dg(y))^T \mathcal{P}_\rho(y),$$

whose norm is at most $M_\rho|x - y|$.

Lemma 3 (Reference construction)**.** *If $\kappa_\rho < 1$, the data-defined map*

$$\mathcal{T}_\rho(u) = \mathcal{K}(\mathcal{P}_\rho(z(u))) \tag{22}$$

is a contraction of U with factor at most κ_ρ. Its unique fixed point defines

$$\bar{u}_\rho = \mathcal{T}_\rho(\bar{u}_\rho), \qquad \bar{x}_\rho = z(\bar{u}_\rho), \qquad \bar{p}_\rho = \mathcal{P}_\rho(\bar{x}_\rho).$$

They satisfy

$$g(\bar{x}_\rho) + B\bar{u}_\rho = 0, \tag{23}$$

$$[\nabla Q(\bar{u}_\rho) + B^T \bar{p}_\rho] \cdot (u - \bar{u}_\rho) \geq 0 \quad (u \in U), \tag{24}$$

$$\nabla L(\bar{x}_\rho) + Dg(\bar{x}_\rho)^T \bar{p}_\rho = \rho \bar{p}_\rho. \tag{25}$$

For an interior reference control, (24) *is the equality $\nabla Q(\bar{u}_\rho) + B^T \bar{p}_\rho = 0$ requested in C61-93.*

Proof. Combine (17), (19), and (21). The product of the three Lipschitz constants is exactly $b^2 M_\rho/[ma(a+\rho)] = \kappa_\rho < 1$. Banach's fixed-point theorem gives the fixed point as the limit of successive iteration of this known map. The displayed equations follow from the definitions. Conversely, any stationary triple whose control minimizes the Hamiltonian satisfies this fixed-point equation, so it coincides with the constructed triple. □

Lemma 4 (Attainment for the full specified family)**.** *For each finite $T > 0$, $x_0 \in X_0$, $\rho \geq 0$, and $\Phi \in \mathcal{F}$, problem* (1) *has an optimizer.*

Proof. All measurable U-valued controls are feasible by Lemma 2. Their states stay in a common compact set and have uniformly bounded derivatives. Take a minimizing sequence. Its controls have a weakly convergent subsequence in $L^2(0,T;\mathbb{R}^r)$, and its states have a uniformly convergent subsequence by equicontinuity. The weak limit is U-valued almost everywhere because this is a closed convex subset of L^2. Passing to the integrated state equation uses uniform convergence of $g(x)$ and weak convergence of Bu against indicator functions of time intervals. Thus the limit is a feasible state-control pair. The state running cost and terminal cost converge. The control integral is weakly lower semicontinuous, since its integrand is continuous and convex in u and the weight is positive. This last fact also follows by taking strongly convergent convex combinations of a weakly convergent sequence and using convexity of Q. The limit therefore attains the infimum. Compactness makes all costs finite, and no convexity of L or Φ was needed. □

4 First variation and the exponential estimate

Lemma 5 (A derived normal current-value adjoint)**.** *For every optimal pair, let p solve the linear backward equation*

$$\dot{p} = (\rho I - Dg(x)^T)p - \nabla L(x), \qquad p(T) = \nabla\Phi(x(T)). \tag{26}$$

Then

$$u(t) = \mathcal{K}(p(t)) \quad \textit{for almost every } t. \tag{27}$$

In particular, this adjoint and its terminal condition are consequences of ordinary first variation; normality and a uniform terminal-gradient bound are not being inferred from bounded terminal values.

Proof. For any measurable U-valued v, the controls $u + \varepsilon(v-u)$ are admissible for $0 \leq \varepsilon \leq 1$, including the state constraint, by Lemma 2. Their state derivative at $\varepsilon = 0$ solves

$$\dot{z} = Dg(x)z + B(v-u), \qquad z(0) = 0.$$

Differentiate the finite-horizon objective and integrate by parts with the explicitly constructed solution of (26). The result is

$$\int_0^T e^{-\rho t}[\nabla Q(u(t)) + B^T p(t)] \cdot [v(t) - u(t)]\, dt \geq 0.$$

Localized variations, first for a countable dense subset of U and then by continuity, imply the corresponding pointwise variational inequality for every $v \in U$, almost everywhere. Strong convexity makes it equivalent to (27). Thus there is no abnormal multiplier issue: the running-cost coefficient was one in a direct derivation from optimality. Strict state invariance justifies the variations without hidden state-boundary measures. Since p is continuous and $\mathcal{K}$ is Lipschitz, the feedback provides a continuous representative of the optimal control. □

Proof of Theorem 1. Existence and the reference were proved above. Write

$$y = x - \bar{x}_\rho, \quad v = u - \bar{u}_\rho, \quad q = p - \bar{p}_\rho.$$

Strong contraction of the drift, the state equation, and (19) give the scalar comparison estimate

$$|y(t)| \le e^{-at}|y(0)| + k\int_0^t e^{-a(t-s)}|q(s)|\,ds. \tag{28}$$

Indeed, the upper derivative of $|y|$ is at most $-a|y| + b|v|$, and $|v| \le (b/m)|q|$.

Subtract (25) from (26). The resulting equation is

$$\dot{q} = (\rho I - Dg(x)^T)q - d(t), \qquad d(t) = \nabla L(x) - \nabla L(\bar{x}_\rho) + (Dg(x) - Dg(\bar{x}_\rho))^T\bar{p}_\rho.$$

By (21), $|d(t)| \le M_\rho|y(t)|$. The backward evolution generated by $\rho I - Dg(x(t))^T$ has norm at most $e^{-(a+\rho)(s-t)}$ from s back to $t \le s$; reverse time and apply (2) to see this. Also $|q(T)| \le H + G/(a+\rho) = H_\rho$. Hence

$$|q(t)| \le H_\rho e^{-(a+\rho)(T-t)} + M_\rho\int_t^T e^{-(a+\rho)(s-t)}|y(s)|\,ds. \tag{29}$$

These estimates have been derived for every optimizer, not assumed for a solution of a separate boundary-value problem.

Set $E_T(t) = e^{-\gamma t} + e^{-\gamma(T-t)}$, and define the finite norms

$$Y = \max_{0\le t\le T}\frac{|y(t)|}{E_T(t)}, \qquad P = \max_{0\le t\le T}\frac{|q(t)|}{E_T(t)}.$$

Elementary integration gives

$$\int_0^t e^{-a(t-s)}E_T(s)\,ds \le \frac{E_T(t)}{a-\gamma},$$
$$\int_t^T e^{-(a+\rho)(s-t)}E_T(s)\,ds \le \frac{E_T(t)}{a+\rho-\gamma}.$$

For example, the two forward summands have the respective bounds $e^{-\gamma t}/(a-\gamma)$ and $e^{-\gamma(T-t)}/(a+\gamma)$; the backward ones are analogous. Since $|y(0)| \le D$, equations (28)–(29) therefore imply

$$Y \le D + \frac{k}{a-\gamma}P, \qquad P \le H_\rho + \frac{M_\rho}{a+\rho-\gamma}Y.$$

The product of the coefficients is $\theta_{\rho,\gamma} < 1$. Solving these two inequalities gives $Y \le A_{\rho,\gamma}$ and $P \le P_{\rho,\gamma}$. Finally $|v| \le (b/m)|q|$ proves (14) with exactly the constant (13).

The admissible range of γ follows by solving $(a-\gamma)(a+\rho-\gamma) > c_\rho$ with $0 < \gamma < a$; its smaller root is (10). To obtain the measure estimate, let $s = \gamma^{-1}\log_+(2C_{\rho,\gamma}/\varepsilon)$. Outside the two endpoint intervals of length s, both exponentials in (14) are at most $\varepsilon/(2C_{\rho,\gamma})$. The exceptional set has ordinary Lebesgue measure at most $2s$, including when the intervals overlap. This argument uses no discounted measure.

If $c_0 < a^2$ and (16) holds, then M_ρ, c_ρ, H_ρ are nonincreasing in ρ, whereas $a+\rho-\gamma$ increases. Consequently

$$\theta_{\rho,\gamma} \le \frac{c_0}{(a-\gamma)^2} < 1, \quad A_{\rho,\gamma} \le A_{0,\gamma}, \quad P_{\rho,\gamma} \le P_{0,\gamma}.$$

This proves the uniform statements; reference continuity is proved next. □

5 Discount dependence of the reference

Proposition 6 (Explicit uniform reference continuity)**.** *Assume $c_0 < a^2$, write $\kappa_0 = c_0/a^2$, and put*

$$L_u = \frac{bG}{ma^2(1-\kappa_0)}, \qquad L_x = \frac{b}{a}L_u, \qquad L_p = \frac{M_0}{a}L_x + \frac{G}{a^2}.$$

For all $\rho, \sigma \geq 0$,

$$|\bar{u}_\rho - \bar{u}_\sigma| \leq L_u|\rho - \sigma|, \quad |\bar{x}_\rho - \bar{x}_\sigma| \leq L_x|\rho - \sigma|, \quad |\bar{p}_\rho - \bar{p}_\sigma| \leq L_p|\rho - \sigma|.$$

In particular the reference converges to the zero-discount reference without any horizon-dependent or terminal-cost-dependent choice.

Proof. The resolvent identity at a fixed x gives

$$|\mathcal{P}_\rho(x) - \mathcal{P}_\sigma(x)| \leq \frac{G|\rho - \sigma|}{(a+\rho)(a+\sigma)}.$$

Compare the fixed points of (22), using one map's contraction bound and then this estimate for the difference between the two maps at a fixed control. Since $\kappa_\rho \leq \kappa_0 < 1$, this yields the asserted L_u. Equation (17) gives L_x, and (21) together with the resolvent identity gives L_p. □

Interpretation of the rate and discount restriction. The bound (10) is a guaranteed admissible rate, not a claim of sharpness or a necessary condition. All of its discount dependence is explicit. Equivalently, writing $k = b^2/m$ and $y = a + \rho$, the small-gain condition is

$$ay^2 - kMy - kKG > 0.$$

This directly specifies a sufficient discount range from the data. For vanishing-discount uniformity the strict margin $c_0 < a^2$ suffices. Failure of that sufficient inequality does not establish absence of a turnpike.

Why the reference generally moves with the discount. For illustration within the theorem, consider scalar $g(x) = -ax$, $B = b$, $L(x) = (x - z)^2/2$, and $Q(u) = mu^2/2$, with a sufficiently large compact control interval and invariant state ball, and $m > b^2/a^2$. The reference is

$$\bar{u}_\rho = \frac{abz}{ma(a+\rho) + b^2}, \qquad \bar{x}_\rho = \frac{b^2 z}{ma(a+\rho) + b^2}.$$

It differs from the undiscounted static minimizer whenever $\rho > 0$ and $bz \neq 0$. This example illustrates the reference formula; the theorem and its proof cover the nonlinear, nonquadratic class above.

6 Why the base assumptions alone are insufficient

Take $X = \{x \in \mathbb{R}^2 : |x| \leq 1\}$, $U = \{0\}$, $L = Q = \Phi = 0$, and

$$\dot{x} = Jx, \qquad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad x(0) = (1/2, 0).$$

All data are smooth, the constraints are compact, the dynamics are uniformly Lipschitz, and for every horizon and discount the unique admissible state-control trajectory is optimal. Terminal costs are uniformly zero. The only stationary pair is $(\bar{x}, \bar{u}) = (0, 0)$. The unique trajectory has $|x(t)| = 1/2$ for every t, so for $\varepsilon = 1/4$ its unweighted exceptional set has measure T. No horizon-independent bound exists. This is a full counterexample to the *base-assumptions-only* implication, which C61-93 itself does not assert.

7 Limits of the claim

- The reference is constructed by a data-defined contraction and obeys the discounted current-value stationary equations (with the correct variational inequality for an active control constraint).
- Optimizer existence and the adjoint are proved; the argument does not start by assuming a suitable canonical-system solution.
- Strict state invariance is required for every control and removes the state-constraint issue in the first-variation derivation.
- The terminal family is uniformly bounded in C^1; bounded values alone do not imply the gradient estimate used in the proof.
- The conclusion is for every optimizer, every positive horizon, all stated initial data and terminal costs, and ordinary Lebesgue measure.
- Both discount-dependent constants and explicit uniformity conditions as $\rho \downarrow 0$ are given.
- The theorem gives sufficient conditions in one class of systems. It is not a classification of every nonlinear drift or a theorem for hard terminal-state constraints.

Use of AI tools

The author used research AI agents that the author built and configured (GPT-6 Astra max, Claude Opus 5.5 max and GPT-6 Astra Pro) during the proof work, for proof search and construction, proof checking, translation, Python computation and similar tasks. The final manuscript was written, and the proofs were checked, by the human author. All responsibility for this work rests with the author personally.

References

[1] T. Faulwasser and L. Grüne, *Turnpike Properties in Optimal Control: An Overview of Discrete-Time and Continuous-Time Results*, arXiv:2011.13670v3, 2021. Sections 2 and 5. `https://arxiv.org/html/2011.13670v3`.